



Lecture 5 – Debugging & Exceptions

30 Exam-Style MCQs (Dr. Ehab Essa Style)

1. Debugging mainly aims to:

- a) Prevent bugs
- b) Find causes
- c) Improve speed
- d) Write tests

Answer: b

2. Debugging is done:

- a) Before coding
- b) After bugs appear
- c) During design
- d) At deployment

Answer: b

3. Which activity identifies the root cause of a bug?

- a) Testing
- b) Debugging
- c) Compilation
- d) Validation

Answer: b

4. A bug is defined as:

- a) Program crash
- b) Incorrect behavior

- c) Syntax error
- d) Exception only

Answer: b

5. Debugging starts after:

- a) Specification
- b) Compilation
- c) Bug discovery
- d) Code review

Answer: c

6. Which is a poor debugging practice?

- a) Reproducing bug
- b) Random changes
- c) Hypothesis testing
- d) Using debugger

Answer: b

7. Print debugging means:

- a) Using IDE
- b) Printing values
- c) Logging errors
- d) Code review

Answer: b

8. Which debugging approach forms a hypothesis first?

- a) Random
- b) Systematic
- c) Trial-and-error
- d) Guessing

Answer: b

9. Which tool allows step-by-step execution?

- a) Compiler
- b) Debugger
- c) Profiler
- d) Linter

Answer: b

10. Changing code before understanding the bug leads to:

- a) Faster fixes
- b) More bugs
- c) Better design
- d) Optimization

Answer: b

11. An exception represents:

- a) Syntax error
- b) Runtime error
- c) Compilation error
- d) Logic flaw

Answer: b

12. Exceptions occur mainly at:

- a) Compile time
- b) Runtime
- c) Design time
- d) Testing time

Answer: b

13. Which Java keyword handles exceptions?

- a) throw
- b) catch
- c) try
- d) All

Answer: d

14. Which keyword is used to signal an exception?

- a) try
- b) catch
- c) throw
- d) finally

Answer: c

15. Code that may cause an exception is placed inside:

- a) catch
- b) finally
- c) try
- d) throws

Answer: c

16. Which block always executes?

- a) try
- b) catch
- c) finally
- d) throw

Answer: c

17. Checked exceptions are checked at:

- a) Runtime
- b) Compile time
- c) Execution
- d) Deployment

Answer: b

18. Unchecked exceptions are subclasses of:

- a) Exception
- b) Error
- c) RuntimeException
- d) Throwable

Answer: c

19. Which is an unchecked exception?

- a) IOException
- b) SQLException
- c) NullPointerException
- d) FileNotFoundException

Answer: c

20. Which exception occurs when dividing by zero?

- a) NullPointerException
- b) ArithmeticException
- c) IOException
- d) AssertionError

Answer: b

21. Exception handling improves:

- a) Performance
- b) Robustness
- c) Speed
- d) Memory

Answer: b

22. Catching exceptions too early may:

- a) Improve clarity
- b) Hide bugs
- c) Stop execution
- d) Improve speed

Answer: b

23. Multiple catch blocks are used to:

- a) Improve speed
- b) Handle different exceptions
- c) Reduce code
- d) Avoid try

Answer: b

24. Which keyword declares exceptions a method may throw?

- a) throw
- b) throws
- c) try
- d) catch

Answer: b

25. Which type should rarely be caught?

- a) Exception
- b) RuntimeException

- c) Error
- d) IOException

Answer: c

26. Assertions are mainly used for:

- a) Input validation
- b) Debugging
- c) Error recovery
- d) Logging

Answer: b

27. An assertion failure indicates:

- a) User error
- b) Programmer error
- c) Hardware error
- d) Runtime crash

Answer: b

28. Exceptions should represent:

- a) Normal flow
- b) Rare conditions
- c) Frequent logic
- d) Loops

Answer: b

29. Good debugging practice includes:

- a) Guessing fixes
- b) Minimal changes
- c) Ignoring logs
- d) Removing tests

Answer: b

30. The best time to add exception handling is:

- a) After crash
- b) During design
- c) During coding
- d) After deployment

Answer: c

Lecture 5 – Debugging & Exceptions

Essay Questions (Dr. Ehab Essa Style)

Essay Question 1

Define debugging. Explain good debugging practices.

Model Answer:

Debugging is the process of **finding and fixing the root cause of bugs** after they are discovered.

It starts **after a bug is observed**, not during testing.

Good debugging practices include:

- Reproducing the bug consistently
- Forming a **hypothesis** about the cause
- Making **small, controlled changes**
- Using debugging tools and logs
- Avoiding random code changes



Essay Question 2

Explain what exceptions are and why exception handling is important.

Model Answer:

An exception is a **runtime error** that disrupts normal program execution.

Exceptions occur when abnormal conditions arise, such as **division by zero** or **null references**.

Exception handling is important because it:

- Prevents program crashes
- Improves **robustness**
- Separates error-handling code from normal logic
- Makes software more reliable



Essay Question 3

Differentiate between checked and unchecked exceptions in Java.

Model Answer:

Checked exceptions are exceptions that are **checked at compile time**.

They must be either **caught or declared** using the **throws** keyword.

Example: **IOException**.

Unchecked exceptions are exceptions that occur at **runtime**.

They are subclasses of **RuntimeException** and do not require explicit handling.

Example: **NullPointerException**.